Mechanical waves and distant effects

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use travel-time reasoning while distinguishing waves from the motion of material.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 14.*
A diverging lens spreads initially parallel rays outward.

02. Worked example: explain the model
Half amplitude; quarter intensity under the stated same-medium model.

03. Read and interpret the model
Disturbance propagates while fluid motion is mostly local oscillation, with possible transport not identical to wave speed.

04. Apply the physics
10 s.

05. Explain your reasoning
Need amplitude/energy, frequency, site conditions, geometry, structures, bathymetry for water waves, etc.

06. Review: connect an earlier idea *Review: Unit 6, Day 5.*
Away from the normal; frequency stays fixed at a stationary boundary.

07. Exit check
Doubles.

08. STRETCH: test the model
Ratio 9: 1; absorption, reflection, terrain, source direction or local coupling.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 14.*
$d_i \approx -10$ cm; $m \approx +1.667$. Upright, enlarged virtual image.

02. Worked example: explain the model
Half amplitude; quarter intensity under the stated same-medium model.

03. Read and interpret the model
Disturbance propagates while fluid motion is mostly local oscillation, with possible transport not identical to wave speed.

04. Apply the physics
P 35 s; S 60 s; delay 25 s.

05. Explain your reasoning
Need amplitude/energy, frequency, site conditions, geometry, structures, bathymetry for water waves, etc.

06. Review: connect an earlier idea *Review: Unit 6, Day 5.*
$\theta_c \approx 36.03°$ from the normal. Above it, total internal reflection occurs.

07. Exit check
31.6 m/s; 316 s; requires long shallow-water wavelength regime and simplified constant depth; not damage forecast.

08. STRETCH: test the model
8/9 W/m²; $P = I\,4\pi r^2 = 3200\pi$ W ≈ 10053 W. Ground-wave geometry, material and attenuation need different models.